

DONGTAI, China

WMO: 582510

Lat: 32.8576N

Lon: 120.2765E

Elev: 13

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.6	25.9	9.1	8.8	31.8	13.2	10.8	31.7	19.6	34.9	17.4	36.6	4.7	290	0.407

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	94.2	81.7	92.1	80.9	89.9	79.9	83.6	90.8	82.6	89.7	81.6	88.1	7.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.9	165.9	87.8	80.8	159.8	87.0	79.7	154.3	86.0	47.8	90.9	46.6	89.9	45.4	88.5	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.0	14.5	12.8	DB	19.9	97.6	2.6	1.9	18.0	99.0	16.5	100.1	15.0	101.2	13.1	102.6
			WB	17.9	85.4	3.1	1.5	15.6	86.5	13.8	87.4	12.0	88.2	9.8	89.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.5	37.0	40.3	48.3	58.6	68.4	75.6	82.3	81.5	73.8	64.3	52.8	41.3
	DBStd	16.64	5.67	6.22	6.60	6.42	5.40	4.58	4.82	4.41	4.58	5.23	6.82	6.25
	HDD50	1127	405	276	112	8	0	0	0	0	0	0	48	278
	HDD65	3494	869	691	518	208	26	0	0	0	1	77	369	735
	CDD50	4945	1	6	59	267	571	768	1002	977	714	442	131	7
	CDD65	1837	0	0	0	17	132	318	537	512	265	54	2	0
	CDH74	16789	0	0	7	150	949	2364	6057	5451	1627	179	5	0
	CDH80	6256	0	0	0	26	275	791	2669	2118	361	16	0	0
	WSAvg	6.1	5.9	6.5	7.1	6.9	6.6	6.3	6.0	6.2	5.5	5.1	5.6	5.7
	PrecAvg	40.8	1.1	1.6	2.1	3.0	3.3	5.5	8.7	6.7	4.8	2.1	2.1	0.9
(15) Precipitation	PrecMax	56.2	2.6	3.9	4.4	7.8	8.5	13.8	28.3	21.0	15.0	7.0	8.0	3.0
	PrecMin	18.2	0.0	0.0	0.5	0.7	0.7	1.7	1.0	0.2	0.8	0.1	0.0	0.0
	PrecStd	8.2	0.6	1.1	1.0	1.8	1.7	3.5	5.9	4.5	3.3	1.8	2.0	0.8
	PrecStd	8.2	0.6	1.1	1.0	1.8	1.7	3.5	5.9	4.5	3.3	1.8	2.0	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.0	63.7	74.2	83.7	91.1	93.7	97.2	96.0	90.9	82.2	73.8	61.7
		MCWB	49.6	54.3	61.2	69.6	70.1	77.3	82.9	83.2	79.0	70.6	62.5	54.4
	2%	DB	51.8	57.8	68.3	78.1	86.6	90.3	94.9	93.3	86.7	78.2	69.3	56.8
		MCWB	46.1	49.4	57.3	65.6	69.5	75.4	82.6	82.3	77.2	66.8	60.6	50.7
	5%	DB	48.4	53.6	63.5	74.4	82.7	87.5	93.1	91.2	83.8	75.7	66.1	53.4
		MCWB	43.5	47.1	54.3	63.4	67.7	74.2	82.0	81.6	75.2	65.8	59.2	48.4
	10%	DB	45.6	50.1	59.5	70.4	79.4	84.7	90.9	89.0	81.5	73.1	63.3	50.8
		MCWB	41.8	45.1	52.4	60.9	66.4	73.3	81.4	80.7	73.7	64.8	57.5	46.8
	0.4%	WB	52.5	57.1	63.2	72.2	76.2	81.4	85.3	85.1	82.4	74.2	67.0	58.2
		MCDB	55.0	62.5	71.1	79.9	83.6	89.4	92.2	92.1	89.1	77.4	69.3	59.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	47.5	51.6	59.2	68.6	73.0	79.7	84.0	83.5	79.9	71.0	63.6	52.3
		MCDB	50.1	56.3	64.9	75.2	79.9	86.6	91.5	90.3	84.2	74.8	66.6	55.2
	5%	WB	44.8	48.7	56.3	65.4	70.9	78.0	82.9	82.5	78.0	68.9	60.8	49.8
		MCDB	47.3	52.0	62.4	71.8	77.7	83.6	90.5	89.0	81.6	72.7	64.5	52.3
	10%	WB	42.4	45.8	53.2	62.4	69.0	76.0	82.0	81.5	76.1	66.7	58.0	47.3
		MCDB	45.2	49.3	58.1	68.4	75.5	81.0	89.2	87.5	79.5	71.0	61.9	50.5
(35) Mean Daily Temperature Range	5% DB	MDBR	12.3	13.1	15.2	16.5	16.5	13.5	11.3	10.9	12.6	14.6	14.4	13.2
		MCDBR	16.7	19.4	22.3	22.9	22.5	18.2	14.3	13.5	14.2	17.0	17.6	16.8
		MCWBR	10.9	12.2	13.0	12.2	9.2	6.5	5.2	5.2	5.8	7.0	9.1	10.6
	5% WB	MCDBR	13.7	16.0	19.8	19.7	17.8	13.7	13.8	12.9	11.5	12.7	13.6	13.1
		MCWBR	10.5	11.5	13.3	12.7	9.6	6.7	5.9	5.5	5.9	7.0	8.9	9.6
(40) Clear-Sky Solar Irradiance	taub		0.575	0.631	0.656	0.660	0.656	0.745	0.743	0.691	0.615	0.632	0.624	0.577
	taud		1.671	1.595	1.567	1.610	1.653	1.531	1.579	1.660	1.770	1.657	1.633	1.681
	Ebn at Noon		189	195	206	214	217	198	197	205	213	192	175	179
	Edh at Noon		63	75	83	83	80	90	86	78	67	70	65	60
	RadAvg		741	869	1223	1456	1536	1402	1476	1443	1237	1040	784	700
(45) All-Sky Solar Radiation	RadStd		88	147	116	161	147	170	189	209	113	140	93	87

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page